Customer Churn Prediction – Final Report

Project: Machine Learning with Scikit-learn **Dataset:** Telco Customer Churn **Model:** K-Nearest Neighbors (KNN)

1. Project Objective

The objective of this project is to predict if a customer will leave a telecommunications company. Customer churn is important because losing customers can reduce company revenue. Machine learning can help a company find customers who may leave and plan actions to keep them.

2. Data and Methodology

The dataset has **7,043 customers** and 21 original columns. I explored the data and checked missing values. I found 11 empty values in **TotalCharges** and filled them using the median calculated from the training data. I removed the customer ID, converted categories into numbers, and scaled the features for KNN.

I used **5,634 customers (80%)** for training and **1,409 (20%)** for testing. I kept a similar churn distribution in both groups. I tested K values of 1, 3, 5, 7, 9, 11 and 15.

3. Model Performance

Among the tested values, **K = 15** had the best accuracy and precision. It also shared the highest recall with K = 9.

Metric	K = 5	K = 15
Accuracy	74.88%	77.29%
Precision	52.72%	57.71%
Recall	51.87%	54.01%

The K = 15 model predicted about **77% of test cases correctly**. However, it identified only about **54% of customers who actually left**. This is useful information, but the model needs improvement before real business use.

4. Key Findings

Contract type: Month-to-month customers had a churn rate of **42.71%**. Customers with two-year contracts had a rate of only **2.83%**.

Payment method: Customers paying by electronic check had a churn rate of **45.29%**, higher than customers using automatic payments.

Internet service: Fiber optic customers had a churn rate of **41.89%**, compared with **18.96%** for DSL customers.

These results show relationships between customer characteristics and churn. They do not prove that these characteristics cause churn.

5. Business Recommendations

I recommend that the company focus on customers with month-to-month contracts and understand why they leave. It should review the electronic check payment experience and investigate prices, service quality and satisfaction for fiber optic customers. The company can test personalized retention campaigns and measure the results before expanding them.

6. Limitations and Future Improvements

The model misses about **46% of customers who leave**. Also, more customers stay than leave, which can affect performance. I compared K values using the test set; in future work, I would choose K with cross-validation on training data and keep the test set for the final evaluation. I would also compare KNN with Logistic Regression and Random Forest.

7. Conclusion

This project showed me how machine learning can support business decisions. KNN can find patterns related to customer churn, but predictions are not perfect. **Good predictions and business understanding need to work together.**